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EMT 202 — Past Questions & Materials
by Renance

Course: EMT 202
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 200

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Paper 1 of 2 — EMT 202, 2024 ENVIRONMENTAL METHOD OF ANALYSIS 11 past question

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

2023/2024 ACADEMIC SESSION SECOND SEMESTER

EMT 222: Environmental method of analysis II (2 Units)

Instruction: Answer 4 questions, at least one from each section 2 hours

1. (a) Why sampling and analysis are an integral part of data quality, between sampling and analysis, which one often generates more errors? Why?

(b) A new method is being developed to analyse a pesticide in soils. A spiked sample with a known concentration of 17.00 mg/kg was measured five times using the new method (data: 15.3, 17.1, 16.7, 15.5, 17.3 mg/kg, $STD_{NEW} = 0.923$) and the established EPA method (data: 15.4, 15.9, 16.7, 16.1, 16.2 mg/kg, $STD_{EPA} = 0.472$). Determine the following parameters for both technique.

(i) Is 15.3mg/kg of the new method data an outlier? *at 95% confidence level*

(ii) Calculate Coefficient of variation

(iii) Relative Standard Deviation

(iv) Variance

(v) Range

(vi) Are the two methods significantly different at 95% confidence level?

(c) Calculate the relative error for both methods. Which one is more accurate and why?

2. (a) Outline four objectives of environmental sampling and analysis

(b) Three laboratories were given a standard reference material (SRM) and were asked to determine its lead concentration in (mg/kg) using the USEPA standard method. Their results were as follows

Laboratory A - 45.2, 47.3, 51.2, 50.4, 52.2, 48.7 ($STD = 2.62$)

Laboratory B - 49.4, 50.3, 51.6, 52.1, 50.9, 51.7 ($STD = 1.01$)

Laboratory C - 48.3, 47.7, 51.6, 55.7, 52.4, 51.5 ($STD = 2.92$)

(i) For laboratory C, is 55.7 mg/kg a possible outlier?

(ii) Calculate the RSD for each laboratory; which laboratory is more precise

- (iii) If the known concentration of lead in SRM is 49.5 mg/kg, calculate the percentage recovery for each laboratory. Which laboratory is more accurate (Note: without considering the deletion of outlier if any)?
- (iv) Are laboratory A & B results significantly different at 95% confidence level?
- (e) The table below has a list of quality control (QC) parameters and the primary means of general QC procedure. Write out the correct letter each corresponding QC procedure. Explain briefly.

QC parameter	Place a letter here	QC procedure
1. Accuracy		a. Analysis of blanks
2. Precision		b. Analysis of matrix spikes
3. Cross-contamination		c. Analysis of replicate samples
4. Extraction efficiency		d. Analysis of reference materials or samples of known concentration

TABLE 7-5

Critical Values for the Rejection Quotient, Q^*

Number of Observations	Q_{crit} (Reject if $Q > Q_{crit}$)		
	90% Confidence	95% Confidence	99% Confidence
3	0.941	0.970	0.994
4	0.765	0.829	0.926
5	0.642	0.710	0.821
6	0.560	0.625	0.740
7	0.507	0.568	0.680
8	0.468	0.526	0.634
9	0.437	0.493	0.598
10	0.412	0.466	0.568

Section 8

- Discuss the importance of measuring pH in environmental studies, focusing on how pH affects water quality, aquatic ecosystems, and the behaviour of pollutants in the environment.
- Explain how alkalinity is related to a water sample's ability to resist pH changes, particularly how it acts as a buffer against acidification and what factors contribute to its acid-neutralizing capacity.
- Describe why measuring acidity is critical in environmental monitoring, focusing on how elevated acidity levels can affect aquatic life, corrode infrastructure, and indicate the presence of pollutants or environmental degradation.

Section C

1.
 - a. State five precautions to take when using an analytical balance.
 - c. Describe the process of calibrating a burette and pipette, highlighting the importance of accuracy and precision in each step.
2.
 - a. List two indicators commonly used in titration, and explain their relevance in determining the end point.
 - b. Differentiate between hard water and soft water, and provide examples of compounds responsible for hardness and softness.
 - c. Outline the step-by-step procedure for determining total hardness in a water sample, including any necessary reagents and equipment.

Paper 2 of 2 — EMT 202, 2023 past question

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- (b) Could you provide a detailed explanation of the steps involved in using an analytical balance for precise measurements? (4 marks).
(c) What factors might increase dissolved oxygen levels in a fish pond? (3marks)

SECTION C

6. (a) Please list four common reagents for determining dissolved oxygen (DO). (4 mark)
(b) What is the significance of alkalinity in the environment, and how does it impact our surroundings? (2marks)
(c) Can you explain the difference between the terms 'titrant' and 'titrand'? (2marks)
(d) What does it signify when the pH of soil contaminated with diesel is measured at 5.9? (2marks)
7. (a) What steps would you take to collect water samples from Agbaroh River for laboratory analysis of physicochemical parameters like pH, while ensuring that the sample's pH is not interfered with? (2 ½ marks)
(b) As an environmental scientist, could you provide a detailed procedure for analysing the pH of a water sample in the laboratory using a pH meter? (5 marks)
(c) Why is it important to calibrate a pH meter before using it to analyze a sample, and what does calibration involve? (2 ½ marks)

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COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

SECOND SEMESTER EXAMINATION FOR 2022/2023 ACADEMIC SESSION
DEGREE EXAMINATION

COURSE TITLE: Methods in Environmental Analysis I

COURSE CODE: EMT 222 (3 UNITS) TIME: 2 hours 30 min

INSTRUCTIONS: Answer four questions, at least one from each section

SECTION A

1. (a) Highlight four uses/applications of titrimetric analysis (4 marks)
 (b) Explain the following (4 marks)
 (i) Equivalent point
 (ii) standard solution
 (c) What are two essential precautions when using an analytical balance? (2 marks)
2. (a) Could you list four indicators commonly used in titration processes? (2 marks)
 (b) Can you identify and describe four types of titrations commonly employed in environmental analysis? (2 marks)
 (c) Could you provide a detailed explanation of one of the titrations mentioned earlier that are used in environmental analysis? (2 marks)
 (d) Can you list four laboratory advantages commonly utilised for titrimetric analysis and provide two schematic diagrams? (4 marks)
3. (a) Comprehensively explain how to separate two immiscible liquids with a relevant example (5 marks)
 (b) Give three examples of immiscible liquid solutions (3 marks)
 (c) Name and sketch a diagram of one laboratory tool used in separating two immiscible liquids (2 marks)

SECTION B

4. (a) What is the concentration of dissolved oxygen (DO) (mg/L) in a sample when 200 mL of the sample requires 8.6 mL of 0.0275N sodium thiosulphate titrant? (5 marks)
 (b) State the significance of dissolved oxygen (DO) to the aquatic environment (5 marks)
5. (a) Could you please clearly and concisely explain the term 'dissolved oxygen'? (DO) (3 marks)

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